

Cloth Buzz



Submitted by:

M.Sharjeel Sarwar 2025-CS-1205

Section C

Supervised by:

Sir.Talha Mubashir

Department of Computer Science

University of Engineering and Technology

Lahore Pakistan

Short Description :

This Project is a console-based Business Application Cloth Buzz is a clothing store management system developed in C++. It allows Admin to and log in, manage clothing inventory, sell articles, generate bills, and track total sales. And users to register and buy articles. The system stores data such as inventory record , passwords of admin and users and stores printed bill in files to ensure persistence and accuracy.

Users of Store : There is a Admin, and users in this application.

Functional requirements :

- User registration and login with password masking.
- Add new clothing articles with ID, name, size, price, and quantity. • View complete inventory search articles by ID.Update stock increase quantity, reduce quantity, update price, sell articles with stock validation
- Automatic bill generation
- Print and save bill details to a file.
- Maintain total sales record.
- Save and load inventory data using files.
- Exit safely with data saved.

Wire Frames :

Welcome Screen

```
***** Cloth Buzz System *****
*****
1. Register User.
2. Login(If you are already registered).
*****
Select option:
```

```
#=====#
#===== Cloth Buzz Login! =====#
#=====#

>Enter Your Username or email: sharjeelsarwar12@gmail.com
>Enter Your Password: *****|
```

```
#=====#  
##### Cloth Buzz #####  
#-----#
```



```
|===== Menu =====|  
1.Add Article.  
2.View Inventory.  
3.Search Article by ID.  
4.Update Stock.  
5.Buy Article.  
6.View Total sales.  
7.Save and Exit..  
|=====|  
Select an Option: |
```

```
#include<iostream>  
#include<fstream>  
#include<conio.h>  
#include<iomanip>  
#include<string>  
using namespace std;  
void Register();  
bool login();  
string getField(string record, int field);  
void readData(string names[], string passwords[], int &indx);  
int Ispresent(string name, string password, string names[], string passwords[],  
int idx);
```

const int Max_ITEMS = 100;

int itemids[Max_ITEMS];

string itemName[Max_ITEMS];

string itemsize[Max_ITEMS];

double itemprice[Max_ITEMS];

int itemQty[Max_ITEMS];

int itemCount = 0;

double Totalsales = 0.0;

int findItem(int id);

bool checkStock(int indx, int qty);

void reduceStock(int indx, int qty);

void loadData();

void saveData();

void Header();

void Menu();

void Itemrow(int indx);

void printBill(string name, string size, int qty, double price, double total);

void Additem();

void ViewInventory();

void Searchitem();

```
void Sellitem();
```

```
void updateStock(int indx);
```

```
int main()
```

```
{
```

```
    system("cls");
```

```
    int ch;
```

```
    bool loggedin = false;
```

```
    while (!loggedin){
```

```
        system("cls");
```

```
        cout<<"\033[33m*===== Cloth Buzz System  
=====*\n\n\033[0m        "<<endl;
```

```
        cout<<"*=====*"<<endl;
```

```
        cout<<"1. Register User."<<endl;
```

```
        cout<<"2. Login(If you are already registered)."<<endl;
```

```
        cout<<"*=====*"<<endl;
```

```
        cout<<"\n\033[34mSelect option:\033[0m ";
```

```
        cin>>ch;
```

```
        if(ch == 1){
```

```
            Register();
```

```
        }
```

```
else if(ch == 2){
    if(login()){
        cout<< "\033[32mLogin Successful Welcome! To Cloth Buzz\033[0m"
        "<<endl;
        loggedin = true;
        cout<< "Press Any key to continue to menu.." <<endl;
        getch();
        break;
    }
    else
    {
        cout<< "\033[31mInvalid Username and Password! (Access
        Denied).\033[0m" <<endl;
        cout<< "Press Any key to Try Again."<<endl;
        getch();

    }
}
else
{
    cout<< "\033[31mInvalid Option.\033[0m"<< endl;
    getch();
}
}
```



```
loadData();
```

```
while(true){  
    Header();  
    Menu();  
    cout<< "\033[34mSelect an Option:\033[0m ";  
    cin>>ch;  
  
    if(ch == 1){  
        Additem();  
        saveData();  
  
    }  
    else if(ch == 2){  
        ViewInventory();  
    }  
    else if(ch == 3){  
        Searchitem();  
    }  
    else if(ch == 4){  
  
    int id;  
    cout << "Enter the ID of the article to update: ";  
    cin >> id;  
  
    int indx = findItem(id);
```

```
if (indx == -1) {
    cout << "Article not found." << endl;
    getch();
} else {
    updateStock(indx);
    cout << "\nUpdated item details:\n";
    Itemrow(indx);
    saveData();
    getch();
}

else if(ch == 5){
    Sellitem();
}

else if(ch == 6){

    cout<< ">Total Sales so far:=====> " << Totalsales << endl;

    cout<< "Press any key to continue..";
    getch();
}

else if(ch == 7){
    saveData();
    cout<< "\033[34mExiting Cloth Buzz... Data saved.\033[0m" << endl;
```



```
        getch();
        break;
    }
    else{
        cout<< "\033[31mInvalid choice. Press any key to try again.\033[0m";
        getch();
    }
}
return 0;
}

string getfield(string record, int field)
{
    int commacount = 1;
    string item = "";
    for(size_t x = 0; x < record.length(); x++)
    {
        if(record[x] == ','){
            commacount = commacount + 1;
            continue;
        }
        else if(commacount == field){
            item = item + record[x];
        }
    }
    return item;
}
```

```
}  
  
void readData(string names[], string passwords[], int &indx)  
{  
    indx = 0;  
    ifstream file("ClothBuzz_items.txt");  
    if(!file.is_open())  
    {  
        return;  
    }  
    string record;  
    while(getline(file, record))  
    {  
        names[indx] = getfield (record, 1);  
        passwords[indx] = getfield(record, 2);  
        indx++;  
  
        if(indx >= 100)  
            break;  
    }  
    file.close();  
}  
  
void Register()  
{  
  
    string user, pass;
```

```
cout<< ">Enter username or email: ";
cin>>user;

cout<< ">Enter Password(include letters,No. or digits): ";
cin>>pass;

ofstream file("Cloth Buzz(pass).txt" , ios:: app);
if(!file.is_open()){
    cout<< "\033[31mError opening file to register user.\033[0m" <<endl;
    return;
}
file << user << "," << pass <<endl;
file.close();

cout<< "\033[32mUser registered successfully!\033[0m" <<endl;

cout<< "Press Any key to continue to login.." <<endl;
getch();
}
bool login()
{

    system("cls");

    cout<<"#=====
=====#" <<endl;
```

```
cout<<"#===== Cloth Buzz Login!
=====#"<<endl;
```

```
cout<<"#=====
=====#\n\n" <<endl;
```

```
string username, password, user, pass;
```

```
cout<< ">Enter Your Username or email: ";
```

```
cin>>username;
```

```
cout<< ">Enter Your Password: ";
```

```
cin>>password;
```

```
ifstream file("Cloth Buzz(pass).txt");
```

```
if(!file.is_open()){
```

```
    return false;
```

```
}
```

```
string record;
```

```
while(getline(file, record))
```

```
{
```

```
    user = getfield(record, 1);
```

```
    pass = getfield(record, 2);
```

```
    if(user == username && pass == password){
```

```
        file.close();
```

```
        return true;
```

```
    }  
}  
file.close();  
return false;  
}  
  
int Ispresent(string name, string password, string names[], string passwords[],  
int idx)  
{  
    int indx = -1;  
    for(int i = 0; i < idx; idx++)  
    {  
        if(name == names[i] && password == passwords[i])  
            indx = i;  
    }  
    return indx;  
}  
  
int findItem(int id){  
    for(int i = 0; i < itemCount; i++){  
        if(itemids[i] == id){  
            return i;  
        }  
    }  
    return -1;  
}  
  
bool checkStock(int indx, int qty){  
    return itemQty[indx] >= qty;  
}
```

```
}
```

```
void reduceStock(int indx, int qty){
```

```
    itemQty[indx] -= qty;
```

```
}
```

```
void loadData(){
```

```
    ifstream file("ClothBuzz_items.txt");
```

```
    if(!file.is_open()){
```

```
        itemCount = 0;
```

```
        Totalsales = 0.0;
```

```
        return;
```

```
    }
```

```
    file >> itemCount;
```

```
    file >> Totalsales;
```

```
    file.ignore();
```

```
    int i = 0;
```

```
    string record;
```

```
    string field;
```

```
    while(getline(file, record))
```

```
    {
```

```
        itemids[i] = stoi(getfield(record, 1));
```

```
        itemName[i] = getfield(record, 2);
```

```
        itemsize[i] = getfield(record, 3);
```

```
        itemprice[i] = stof(getfield(record, 4));
```

```
        itemQty[i] = stof(getfield(record, 5));
        i++;
        if(i >= 100)
            break;
    }
    file.close();
}

void saveData(){
    ofstream file("ClothBuzz_items.txt");
    if(!file.is_open()){
        cout<< "Error Saving data!" << endl;

        return;
    }
    file << itemCount << endl;
    file << Totalsales << endl;

    for(int i = 0; i < itemCount; i++){
        file << itemids[i] << "," << itemName[i] << "," << itemsize[i] << "," <<
itemprice[i] << "," << itemQty[i] << endl;
    }
    file.close();
}

void Header() {
    system("cls");
```

[illegible]

```
void Menu(){
```

```
cout<<"|===== Menu =====|"<<endl;
```

```
cout << "1.Add Article." << endl;
cout << "2.View Inventory." << endl;
cout << "3.Search Article by ID." << endl;
cout << "4.Update Stock." << endl;
cout << "5.Buy Article." << endl;
cout << "6.View Total sales." << endl;
cout << "7.Save and Exit.." << endl;

cout<<"|=====|"<<endl;
}
```

```
void Itemrow(int indx){
```



```
cout << "ID: " << itemids[indx]
    << "| Name: " << itemName[indx]
    << "| Size: " << itemsize[indx]
    << "| Price: " << itemprice[indx]
    << "| Stock: " << itemQty[indx] << endl;
}

void printBill(string name, string size, int qty, double price, double total){

    time_t now = time(0);
    char*dt = ctime(&now);

    cout << "\n\033[32m*-----Total Bill-----
*\033[0m" << endl;

    cout << "Date & Time: " << dt;
    cout << "Item: " << name << " (" << size << ")" << endl;
    cout << "Quantity: " << qty << endl;
    cout << "Price per Item: " << price << "Rs" << endl;
    cout << "Total Amount: " << total << "Rs" << endl;
    cout << "\033[32m*-----*\033[0m" << endl;

    cout << "\033[33mThank You For Visiting Cloth Buzz! Happy
Shoping.\033[0m" << endl;

    ofstream bout("Bill.txt", ios::app);
    if(!bout.is_open()){
        cout << "\033[31mError Could not open Bill.txt to print Bill.\033[0m" <<
endl;
```

```
    getch();
    return;
}

bout<<"*-----Bill-----*" << endl;
bout << "Date & Time: " << dt;
bout << "Item: " << name << " (" << size << ")" << endl;
bout << "Quantity: " << qty << endl;
bout << "Price per Item: " << price << "Rs" << endl;
bout << "Total Amount: " << total << "Rs" << endl;
bout <<"*-----*" <<endl << endl;

bout << "Thank You For Visiting Cloth Buzz! Happy Shopping." << endl;

    bout.close();

    cout << "\n\033[1mBill has also been printed to Bills.txt\033[0m" << endl;
    getch();

}

void Additem(){
    if(itemCount >= Max_ITEMS){
        cout<< "\033[34mCannot add more items inventory full.\033[0m" <<
endl;
        getch();
        return;
    }
}
```

```
int id;
string name, size;
double price;
int qty;

cout << "Enter Article's ID: ";
cin>>id;

if(findItem(id) != -1){
    cout << "Item ID already exists. Try another ID." << endl;
    cout << "Press any key to continue..";
    getch();
    return;
}

cout << "Enter Article Name: " ;
cin.ignore();
getline(cin, name);
cout << "Enter Article size (S/M/L/XL): ";
getline(cin, size);
cout << "Enter Price of article: ";
cin>> price;
cout << "Enter Quantity: ";
```

```
    cin>>qty;

    itemids[itemCount] = id;
    itemName[itemCount] = name;
    itemsize[itemCount] = size;
    itemprice[itemCount] = price;
    itemQty[itemCount] = qty;

    itemCount++;

    cout << "\033[32mNew Article Added successfully.\033[0m" << endl;
    getch();

}

void ViewInventory(){
    if (itemCount == 0){
        cout << "No articles in inventory." << endl;
        getch();
        return;
    }
    ifstream fin("ClothBuzz_items.txt" , ios::in);
    if(!fin.is_open()){
        cout << "Error opening File.";
        getch();
        return;
    }
}
```

```
}

cout << "\n\033[33m*=====--INVENTORY--
=====*\033[0m" << endl;

for(int i = 0; i < itemCount; i++){
    Itemrow(i);

}

cout <<
"\033[33m*=====
=====*\033[0m" <<endl;

getch();
fin.close();
}

void updateStock(int indx){
    int op, qty;
    float nprice;
    cout <<"===== "<< endl;
    cout << "1.Add Stock."<< endl;
    cout << "2.Reduce Stock."<< endl;
    cout << "3.Update price."<< endl;
    cout <<"===== "<< endl;

    cout<< "\033[34mEnter Choice:\033[0m" << endl;
    cin>>op;

    if(op == 1){
```

```
    cout << "Enter Quantity to Add: ";
    cin>>qty;
    itemQty[indx] += qty;
    cout <<"\033[32mStock Added Successfully.\033[0m" << endl;
}
else if(op == 2){
    cout << "Enter Quantity to Reduce: ";
    cin>>qty;

    if (itemQty[indx] >= qty ){
        itemQty[indx] -= qty;
        cout <<"\033[32mStock Updated Successfully.\033[0m" << endl;
    }
    else {
        cout <<"\033[31mNot Enough Stock.\033[0m" << endl;
    }

}
else if(op == 3){
    cout << "Enter New Price: ";
    cin>>nprice;

    if(nprice > 0){
        itemprice[indx] = nprice;
```

```
        cout << "\033[32mPrice Updated Successfully!\033[0m" << endl;

    }

    else {
        cout << "\033[31mInvalid price\033[0m" << endl;
    }
}

else {
    cout << "\033[31mInvalid Choice.\033[0m" << endl;
}
}

void Searchitem(){
    int id;

    cout << "Enter the ID to search: ";
    cin>>id;

    int indx = findItem(id);
    if(indx == -1){
        cout << "Article not found." << endl;

    }
    else {
        cout << "\033[32mArticle Found.\033[0m" << endl;
```

```
        Itemrow(indx);

    }
    getch();
}

void Sellitem(){
    int id;
    cout << "Enter article ID to Buy: ";
    cin>> id;

    int indx = findItem(id);
    if(indx == -1){
        cout << "Article not found!" << endl;
        getch();
        return;
    }
    int qty;
    cout << "Enter the Quantity to sell:";
    cin>> qty;

    if(qty <= 0){
        cout << "\033[31mInvalid quantity.\033[0m" << endl;
        getch();
        return;
    }
}
```



```
    if(!checkStock(indx, qty)){  
        cout << "\033[31mInsufficient stock.Available:\033[0m " <<  
itemQty[indx] << endl;  
        getch();  
        return;  
    }  
    double billAmount = qty * itemprice[indx];  
    reduceStock(indx, qty);  
    Totalsales += billAmount;  
  
    printBill(itemName[indx], itemsize[indx], qty, itemprice[indx],  
billAmount);  
    getch();  
}
```

Limitations:

- Console-based (no actual graphics)
- Passwords stored in plain text (not encrypted).
- Fixed maximum number of items.
- No graphical bill printing (text-only).

Future Enhancements: Can be made more realistic business store by enabling (GUI) graphics , password should be encrypted. Can be added barcode support.

